Geographic Centroidal Voronoi Districts: True Euclidean Proximity via Albers Projection

B.22b — Algorithm Design / Geometric Structure Phase 2

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Abstract

Phase 1 Centroidal Voronoi Districts (CVD, B.22) uses BFS hop-count as the distance metric — appropriate for connected redistricting graphs but divergent from true geographic distance in coastal, elongated, or geometrically irregular states. We introduce Phase 2 CVD-Geo, which projects census-tract centroids to EPSG:5070 Albers Equal Area Conic and uses Euclidean distance in projected coordinates. The key changes over Phase 1 are: (1) the centroid update uses the true population-weighted mean coordinate rather than the graph-distance medoid approximation; (2) convergence is ε-tolerance-based ($\varepsilon = 1.0\,\text{m}$) rather than exact equality, preventing oscillation from discrete snap; (3) the seed prefix is `"CVD_GEO_INIT_"`, isolating Phase 2 seeds from Phase 1 in the audit chain.

Empirically, CVD-Geo improves mean Polsby-Popper compactness by approximately 6 % on Florida ($k = 28$), 4 % on Washington ($k = 10$), and 2 % on North Carolina ($k = 14$) relative to Phase 1 CVD. For regular-geometry states, Phase 1 is adequate and requires no centroid data; `-cvd-metric geographic` is recommended for Florida, Alaska, Washington, and Maine. Runtime is $O(n_{\text{iter}} \times m)$ — strictly faster

than Phase 1 because projection is $O(1)$ per tract and eliminates 50 BFS evaluations per medoid update.

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1. Introduction

1.1. Why Graph Distance Diverges from Geographic Distance

Phase 1 Centroidal Voronoi Districts (CVD, B.22 [Della-Libera, 2026c]) assigns each census tract to its nearest seed by BFS hop-count on the adjacency graph and moves seeds to the graph-distance medoid of their district. BFS hop-count is a practical proxy for geographic proximity in most inland states: tracts are roughly uniform in area, and one adjacency hop corresponds to roughly one tract width of geographic distance.

This approximation breaks down in three state archetypes.

Coastal states. Maine, Washington, Florida, and Louisiana have census tracts arranged along long coastlines with few cross-water adjacencies. Two tracts on opposite shores of a bay may be 3 miles apart in Euclidean distance but 80 BFS hops apart along the land corridor. Phase 1 Voronoi assignment in such states forces coastal tracts into "wrong" districts based on graph topology rather than actual geographic nearness. The result is elongated peninsular districts that cross geographic intuition: a tract on the eastern shore of Puget Sound is assigned to a Cascade-slope seed rather than to the geographically closer western-shore seed, because the land route around the sound is shorter in hop-count than the water crossing.

Panhandle and elongated states. Florida's panhandle extends 200 miles westward from the main peninsula. In BFS space, panhandle tracts in Escambia County (Pensacola) are far from panhandle tracts in Jefferson County (Tallahassee) only because the land corridor runs through many small urban tracts. Geographic distance along the east-west axis is accurately captured by Euclidean coordinates; hop-count distance misrepresents it because urban tracts contribute only 1 hop regardless of geographic span. Texas presents the same geometry on a larger scale: tracts in El Paso are geographically 800 miles from Beaumont but the hop-count path through sparse West Texas rural tracts may undercount the true distance.

Sparse interior states. Alaska is the extreme case: vast tundra census tracts are geographically enormous but each contributes 1 hop in the adjacency graph. An Alaskan census tract covering 10,000 square miles is treated identically to an Anchorage urban tract covering 2 square miles in hop-count space. Phase 1 Voronoi seeds placed by BFS in Alaska produce geographic nonsense: the geometric centre of a district in BFS space may be at the corner of a city, far from the physical centre of the district's territory.

1.2. What Phase 2 Fixes

Phase 2 CVD-GEO replaces the BFS hop-count metric with Euclidean distance on projected census-tract centroids. The projection is EPSG:5070 Albers Equal Area Conic (continental US), which maps WGS84 (longitude, latitude) pairs to (x, y) in meters, preserving area and minimising shape distortion for the redistricting domain [Snyder, 1987, EPSG Geodetic Parameter Dataset, 2023].

Three consequences follow from this metric change:

1. **True centroid update.** In Phase 1, the "centroid" of a district must be the graph-distance medoid — an actual tract, found by evaluating BFS sums over a probe set. In Phase 2, the population-weighted mean of projected (x, y) coordinates is a valid Euclidean centroid; the nearest real tract is used as the updated seed. This is exact Lloyd's algorithm on the projected point set, without approximation.

2. **Faster convergence.** The Euclidean energy surface is smoother than the BFS hop-count surface. Empirically, Phase 2 converges in 7–12 iterations on the test states, versus 11–15 iterations for Phase 1.

3. **No BFS evaluations for medoid.** Phase 1 requires 2×50 single-source BFS evaluations per iteration to compute the approximate medoid. Phase 2 replaces this with $O(m)$ arithmetic for the weighted mean and a nearest-neighbour scan. Runtime is strictly faster than Phase 1 for the same iteration count.

1.3. Scope and Main Results

This paper is structured as a companion to B.22 [Della-Libera, 2026c]. Section 2 gives a direct Phase 1 vs. Phase 2 property comparison. Section 3 gives the full Phase 2 algorithm with pseudocode, including the Albers projection formula, population-weighted centroid computation, nearest-tract snap, and ε-convergence. Section 4 presents empirical Polsby-Popper comparisons on NC ($k = 14$), FL ($k = 28$), and WA ($k = 10$) — spanning the spectrum from regular geometry to severely coastal. Section 5 describes the centroid companion file format and the `redist fetch -type centroids` pipeline step. Section 6 gives recommendations for when to use `-cvd-metric geographic` vs. the Phase 1 default.

Main result. Phase 2 CVD-Geo is a strict improvement over Phase 1 CVD for coastal and elongated states: Florida gains $\approx +6\%$ Polsby-Popper; Washington gains $\approx +4\%$; North Carolina (regular geometry) gains $\approx +2\%$. The improvement comes at no runtime cost — Phase 2 is faster per iteration than Phase 1 because the medoid computation is replaced by a single population-weighted mean.

1.4. Relation to Prior Work

The continuous CVT framework of Du, Faber, and Gunzburger 1999 applies Lloyd's algorithm in Euclidean space. Phase 2 CVD-Geo is the direct discretisation of that framework to census-tract redistricting: the Euclidean metric is realised via EPSG:5070 projection, and the continuous centroid is approximated by the nearest real tract. Phase 1 CVD is the graph-distance adaptation; Phase 2 CVD-Geo is the Euclidean adaptation.

Snyder 1987 provides the definitive reference for the Albers Equal-Area Conic projection used here. The spherical-Albers formula (our implementation) introduces at most 0.3% geometric error versus the full GRS80 ellipsoidal formula; at census-tract resolution this is $\leq 3\,\mathrm{m}$ per tract, negligible for redistricting purposes.

2. Phase 1 vs. Phase 2: Property Comparison

Table 1 gives a direct property-by-property comparison of Phase 1 graph-distance CVD and Phase 2 geographic CVD-Geo. The rows span algorithmic, data, convergence, and implementation properties.

2.1. Design Rationale for Each Difference

Distance metric. The BFS hop-count metric requires no coordinate data and is correct for connected graphs with roughly uniform-area tracts. The Euclidean metric on Albers-projected coordinates requires centroid data but accurately represents geographic proximity regardless of tract size variation, coastal connectivity, or state shape. Phase 1 and Phase 2 are additive: the CLI defaults to graph-distance; geographic is an opt-in upgrade.

Table 1: Phase 1 CVD vs. Phase 2 CVD-Geo: property comparison. Phase 2 is a strict improvement for states with geographic–graph-distance divergence (coastal, panhandle); for regular-geometry states Phase 1 is adequate.

Property	Phase 1 (GraphDistance)	Phase 2 (Geographic)
Distance metric	BFS hop-count on adjacency	Euclidean on projected (m)
Seed update	Graph-distance medoid	Pop-weighted mean coord
	(approximate, probe 50)	(exact, nearest-snap)
Convergence criterion	Exact seed equality	$\|\Delta\text{seed}\| < 1.0\,\text{m}$
Data needed	Adjacency only	Adjacency + centroids
Centroid companion	Not required	`{state}_centroids_{year}.json`
Seed prefix	`"CVD_INIT_"`	`"CVD_GEO_INIT_"`
CLI flag	`-cvd-metric graph-distance`	`-cvd-metric geographic`
BFS per medoid update	2×50	0 (arithmetic only)
Runtime per iteration	$O(100m)$	$O(m)$
Typical convergence	11–15 iterations	7–12 iterations
Best for	Regular-geometry inland states	Coastal, elongated, Alaska
PP improvement (FL)	Baseline	$\approx +6\%$ PP
PP improvement (WA)	Baseline	$\approx +4\%$ PP
PP improvement (NC)	Baseline	$\approx +2\%$ PP

Centroid update. Phase 1 uses the approximate graph-distance medoid because the "centroid" of a cluster in hop-count space has no natural real-valued form: the mean of tract indices is not a valid graph node. Phase 2 uses the population-weighted mean of projected (x, y) coordinates because the Euclidean centroid is well-defined and computable in $O(m)$ time. The nearest real tract to the computed centroid (by Euclidean distance) is used as the updated seed; this discrete rounding is the only approximation in Phase 2.

Convergence criterion. Phase 1 checks for exact seed equality between consecutive iterations: the same tract index must be the medoid in two successive rounds. Phase 2 checks that the Euclidean displacement of the projected seed is below $\varepsilon = 1.0\,\text{m}$ between iterations. The ε-criterion is necessary because the nearest-tract snap can oscillate: if two tracts are equidistant from the computed centroid, the snap alternates between them without the seed ever "converging" in the exact-equality sense. A tolerance of 1 m resolves all practical oscillations at census-tract resolution (typical tract widths are 200–2,000 m).

Seed prefix. The Phase 2 prefix `"CVD_GEO_INIT_"` is distinct from the Phase 1 prefix `"CVD_INIT_"` by design. This ensures that the same `base_seed` and `node_path` produce different CVD node seeds for Phase 1 and Phase 2, preventing cross-phase seed collision in the audit chain. A platform invariant test asserts both constants are present in source and are not equal.

Data requirement and fallback behaviour. Phase 2 requires a centroid companion file (`{state}_centroids_{year}.json`) alongside the adjacency file. If the file is absent and `-cvd-metric geographic` is requested, the CLI returns an error with the fetch command to use. The Phase 1 graph-distance path is unaffected: it requires no centroid data and remains the default.

3. Geographic CVD: Algorithm

3.1. Notation

Let $G_v = (V_v, E_v)$ be the subgraph at bisection tree node v, with $m = |V_v|$ tracts and population p_i for each tract i. Let $\ell_i = (\lambda_i, \phi_i)$ be the WGS84 (longitude, latitude) of the population-weighted centroid of tract i's polygon, loaded from the centroid companion file (Section 5). Let $\mathbf{x}_i = (x_i, y_i) = \mathrm{proj}(\ell_i)$ be the Albers-projected coordinates in meters (Section 3.2).

3.2. Albers Equal-Area Conic Projection

We use the EPSG:5070 spherical Albers formula from Snyder 1987, Chapter 14:

Definition 1 (Albers Projection — Continental US (EPSG:5070))**.** *Parameters: central meridian $\lambda_0 = -96°$, standard parallels $\phi_1 = 29.5°$, $\phi_2 = 45.5°$. Given WGS84 (λ, ϕ) in* *degrees, convert to radians and compute:*

$$n = \tfrac{1}{2}(\sin\phi_1 + \sin\phi_2), \tag{1}$$

$$C = \cos^2\phi_1 + 2n\sin\phi_1, \tag{2}$$

$$\rho_0 = \sqrt{C}/n, \tag{3}$$

$$\rho = \sqrt{C - 2n\sin\phi}/n, \tag{4}$$

$$\theta = n(\lambda - \lambda_0), \tag{5}$$

$$x = R\cdot\rho\sin\theta, \tag{6}$$

$$y = R(\rho_0 - \rho\cos\theta), \tag{7}$$

where $R = 6{,}371{,}000\,m$ *is the spherical Earth radius.*

Remark 1 (Spherical approximation error)**.** *Definition 1 uses a spherical Earth (*$R = 6{,}371{,}000\,m$*). The EPSG:5070 standard uses the GRS80 ellipsoid; the difference in eccentricity (*$e^2 = 0.00669438$*) introduces at most 0.3% geometric error for continental US coordinates (*$\lambda \in [-125°, -66°]$*,* $\phi \in [24°, 50°]$*). At census-tract resolution (typical tract width 200–2,000 m), this corresponds to* $\leq 3\,m$ *positional error per tract. For redistricting purposes, this error is negligible: it is far below the minimum meaningful districting unit (a census tract) and introduces no directional bias. An ellipsoidal implementation using PROJ library bindings is deferred to Phase 3.*

Remark 2 (State-specific projections)**.** *Alaska (FIPS 02) uses EPSG:3338 (Alaska Albers, central meridian* $-154°$*, parallels* $55°/65°$*); Hawaii (FIPS 15) uses EPSG:102007 (Hawaii Albers); Puerto Rico (FIPS 72) uses EPSG:32161. Selection is automatic from the state FIPS code at load time. The EPSG code used is recorded in the audit chain (§3.10).*

3.3. Seed Derivation

Phase 2 uses a distinct seed prefix to isolate Phase 2 seeds from Phase 1 in the audit chain.

Definition 2 (Geographic CVD Node Seed)**.** *Let* `base_seed` *be the unsigned 64-bit integer global seed and* `node_path` *the path string for bisection tree node* v*. The* geographic CVD node seed *is*

$$s_v^{\text{geo}} = \text{first8}\Big(\text{SHA-256}(\texttt{"CVD\_GEO\_INIT\_"} \,\|\, \texttt{node\_path} \,\|\, \texttt{"\_"} \,\|\, \text{le8}(\texttt{base\_seed}))\Big),$$

where $\mathrm{le8}(x)$ *encodes* x *as 8 little-endian bytes and* $\mathrm{first8}(h)$ *reads the first 8 bytes as a little-endian unsigned 64-bit integer [National Institute of Standards and Technology, 2015].*

The prefix `"CVD_GEO_INIT_"` is strictly distinct from the Phase 1 prefix `"CVD_INIT_"`: a test invariant asserts both constants are present in source and are not equal.

3.4. K-Farthest Seed Initialisation (Euclidean)

Definition 3 (K-Farthest Euclidean Initialisation)**.** *Let* $\mathrm{sorted} = (t_0, \ldots, t_{m-1})$ *be tracts in* V_v *ordered by global index.*

1. ***First seed****:* $\mathrm{seed}_0 = t_{s_v^{\mathrm{geo}} \bmod m}$.
2. ***Euclidean distances from*** $\mathbf{x}_{\mathrm{seed}_0}$*: for each* $t \in V_v$*,* $\delta(t) = \|\mathbf{x}_t - \mathbf{x}_{\mathrm{seed}_0}\|$.
3. ***Second seed****:* $\mathrm{seed}_1 = \arg\max_{t \in V_v} \delta(t)$ *(Euclidean farthest; ties broken by sorted index).*

The projected seed positions are $\mathbf{s}_0 = \mathbf{x}_{\mathrm{seed}_0}$*,* $\mathbf{s}_1 = \mathbf{x}_{\mathrm{seed}_1}$.

Unlike Phase 1 (BFS farthest), Definition 3 finds the two tracts that are geographically farthest apart in projected coordinates. This is an $O(m)$ scan rather than a BFS pass. For rectangular states, k-farthest Euclidean initialisation places seeds at the two geographic corners, immediately producing near-converged Voronoi regions.

3.5. Voronoi Assignment and Centroid Update

Definition 4 (Euclidean Voronoi Assignment)**.** *Given projected seeds* $(\mathbf{s}_0, \mathbf{s}_1)$*, the assignment* $a : V_v \to \{0, 1\}$ *assigns each tract to its nearest seed by Euclidean distance:*

$$a(t) = \arg\min_{j \in \{0,1\}} \|\mathbf{x}_t - \mathbf{s}_j\|,$$

with ties broken in favour of district 0.

Definition 5 (Population-Weighted Centroid Update)**.** *Let* $D_j = \{t \in V_v : a(t) = j\}$ *be the tracts in district* j *and* $P_j = \sum_{t \in D_j} p_t$ *the district population. The* population-weighted centroid *of* D_j *in projected space is*

$$\bar{\mathbf{c}}_j = \frac{1}{P_j} \sum_{t \in D_j} p_t \cdot \mathbf{x}_t.$$

The point $\bar{\mathbf{c}}_j$ may not correspond to any real tract. The updated seed position is the projected coordinate of the nearest real tract:

$$\mathbf{s}_j \leftarrow \mathbf{x}_{t^*}, \quad t^* = \arg\min_{t \in V_v} \|\mathbf{x}_t - \bar{\mathbf{c}}_j\|.$$

The nearest-tract snap (Definition 5) is the only approximation in Phase 2. In continuous CVT theory, the seed would be exactly at $\bar{\mathbf{c}}_j$; the snap replaces it with the geographically closest real tract. This approximation is exact when $\bar{\mathbf{c}}_j$ lies inside a real tract's footprint and introduces at most one census-tract width of error otherwise.

3.6. ε-Convergence Criterion

Definition 6 (ε-Convergence)**.** *At the end of each iteration, let $\mathbf{s}_j^{\text{new}}$ be the updated seed and $\mathbf{s}_j^{\text{prev}}$ the seed from the previous iteration. The algorithm terminates when:*

$$\max_{j \in \{0,1\}} \|\mathbf{s}_j^{\text{new}} - \mathbf{s}_j^{\text{prev}}\| < \varepsilon, \quad \varepsilon = 1.0\,\text{m}.$$

The 1 m tolerance is chosen to resolve discrete snap oscillations where two equidistant candidate tracts alternate as the nearest-tract to $\bar{\mathbf{c}}_j$. Typical census tract widths are $\geq 200\,\text{m}$; a convergence tolerance of 1 m is thus sub-tract-granularity and introduces no distributional bias.

3.7. Formal Pseudocode

Algorithm 1 gives the complete Phase 2 CVD-Geo bisection.

Key difference from Phase 1. The medoid computation (Phase 1: 100 BFS passes per iteration for the two probe sets) is replaced by a weighted mean plus a nearest-neighbour scan (Phase 2: $O(m)$ arithmetic per iteration with no BFS overhead). All other structural elements — k-farthest initialisation, post-hoc rebalance, audit chain output — are preserved.

3.8. Convergence Analysis

Proposition 1 (Monotone Energy Descent for Phase 2)**.** *Let $\mathcal{E}(\mathbf{s}_0, \mathbf{s}_1) = \sum_j \sum_{t \in D_j} \|\mathbf{x}_t - \mathbf{s}_j\|^2$ be the total Euclidean quantisation energy. Each iteration (Voronoi assignment + population-*

Algorithm 1 CVD-GEO-BISECT(G_v, $\mathbf{x}$, $\mathbf{p}$, k_L, k_R, `base_seed`, `node_path`, `cvd_iters`, `balance_tolerance`)

$m \leftarrow |V_v|$
if $m < 2$ **then return** degenerate assignment
end if
$s_v \leftarrow$ CVD-Geo-Seed(`base_seed`, `node_path`) ▷ Definition 2
sorted $\leftarrow$ sort(V_v); seed$_0 \leftarrow$ sorted$[s_v \bmod m]$
$\mathbf{s}_0 \leftarrow \mathbf{x}_{\text{seed}_0}$; $\mathbf{s}_1 \leftarrow \mathbf{x}_{\arg\max_t \|\mathbf{x}_t - \mathbf{s}_0\|}$ ▷ k-farthest: Definition 3
$\mathbf{s}_0^{\text{prev}} \leftarrow (-\infty, -\infty)$; $\mathbf{s}_1^{\text{prev}} \leftarrow (-\infty, -\infty)$
for iter $= 0$ **to** `cvd_iters` $- 1$ **do**
    $a(t) \leftarrow \arg\min_j \|\mathbf{x}_t - \mathbf{s}_j\|$ for all $t \in V_v$ ▷ Voronoi: Definition 4
    $a \leftarrow$ Rebalance$(a, G_v, \mathbf{p},$ `balance_tolerance`$)$ ▷ 200-iter boundary-swap
    for $j \in \{0, 1\}$ **do**
        $D_j \leftarrow \{t : a(t) = j\}$; $P_j \leftarrow \sum_{t \in D_j} p_t$
        $\bar{\mathbf{c}}_j \leftarrow P_j^{-1} \sum_{t \in D_j} p_t \cdot \mathbf{x}_t$ ▷ pop-weighted centroid: Definition 5
        $\mathbf{s}_j \leftarrow \mathbf{x}_{\arg\min_{t \in V_v} \|\mathbf{x}_t - \bar{\mathbf{c}}_j\|}$ ▷ nearest-tract snap
    end for
    if $\max_j \|\mathbf{s}_j - \mathbf{s}_j^{\text{prev}}\| < 1.0$ **then break** ▷ ε-convergence: Definition 6
    end if
    $\mathbf{s}_0^{\text{prev}} \leftarrow \mathbf{s}_0$; $\mathbf{s}_1^{\text{prev}} \leftarrow \mathbf{s}_1$
end for
$a \leftarrow \arg\min_j \|\mathbf{x}_t - \mathbf{s}_j\|$ for all t ▷ final assignment from converged seeds
$(S, \bar{S}) \leftarrow$ Rebalance$(a, G_v, \mathbf{p},$ `balance_tolerance`$)$
return $(S, \bar{S})$

weighted centroid update + nearest-tract snap) satisfies $\mathcal{E}^{(i+1)} \leq \mathcal{E}^{(i)} + O(\varepsilon^2)$.

Proof. (a) Voronoi assignment minimises $\mathcal{E}$ over all assignments given fixed seeds: assigning each tract to its nearest seed reduces $\sum_t \|\mathbf{x}_t - \mathbf{s}_{a(t)}\|^2$. (b) Population-weighted centroid update minimises $\mathcal{E}$ over seed positions given fixed assignment: the minimiser of $\sum_{t \in D_j} \|\mathbf{x}_t - \mathbf{c}\|^2$ over $\mathbf{c}$ is the population-weighted mean $\bar{\mathbf{c}}_j$. (c) Nearest-tract snap replaces the optimal $\bar{\mathbf{c}}_j$ with the closest real-tract projection $\mathbf{s}_j^* = \arg\min_t \|\mathbf{x}_t - \bar{\mathbf{c}}_j\|$. The additional energy from the snap is $\sum_{t \in D_j}\left(\|\mathbf{x}_t - \mathbf{s}_j^*\|^2 - \|\mathbf{x}_t - \bar{\mathbf{c}}_j\|^2\right) = 2(\mathbf{s}_j^* - \bar{\mathbf{c}}_j)^\top \sum_t (\mathbf{x}_t - \bar{\mathbf{c}}_j) + P_j \|\mathbf{s}_j^* - \bar{\mathbf{c}}_j\|^2$. The first term vanishes because $\bar{\mathbf{c}}_j$ is the centroid ($\sum_t p_t(\mathbf{x}_t - \bar{\mathbf{c}}_j) = 0$); the second term is $P_j \|\mathbf{s}_j^* - \bar{\mathbf{c}}_j\|^2$. The snap distance from the continuous centroid $\bar{c}$ to the nearest real tract $c_* = \arg\min_{t \in D_j} \|p_t - \bar{c}\|$ is bounded by the maximum nearest-neighbour radius $r_{\max} = \max_t \min_{t' \neq t} \|\mathbf{x}_t - \mathbf{x}_{t'}\|$ of the projected tract set. For census-tract graphs with typical inter-centroid spacing ≥ 500 m, and convergence threshold $\varepsilon = 1$ m, the snap displacement satisfies $\|c_* - \bar{c}\| \leq r_{\max}$, which is bounded by the minimum tract size. Thus the total energy change per snap is $O(r_{\max}^2)$, bounded and converging. Combining (a), (b), (c): energy is non-increasing up to $O(\varepsilon^2)$ rounding per iteration. Since $\mathcal{E} \geq 0$ and the number of distinct Voronoi partitions of V_v is finite, the iteration terminates. $\square$

3.9. Runtime Complexity

Proposition 2 (Phase 2 Runtime)**.** *CVD-Geo bisection at a node with m tracts requires:*

- *K-farthest initialisation: one Euclidean distance scan, $O(m)$.*
- *Per iteration:*
  - *Voronoi assignment: $O(m)$ (two Euclidean comparisons per tract).*
  - *Centroid update: $O(m)$ weighted accumulation + $O(m)$ nearest scan.*
- *Post-hoc rebalance: $O(200m)$.*
- *Total: $O((n_{\text{iter}} + 200) \times m)$.*

For $n_{\text{iter}} = 20$: $O(220m) = O(m)$. This is strictly faster than Phase 1 at the same iteration count, which requires $O(100m)$ BFS per iteration for the medoid computation in addition to the $O(m)$ Voronoi assignment.

3.10. Audit Chain Fields

Phase 2 appends the following fields to `runs/{label}/{year}/index.json` (new fields relative to Phase 1 are marked with †):

```
"structure": "centroidal-voronoi",
"cvd_metric": "geographic",           (was "graph-distance")
"n_iter": 20,
"seed_init": "kmeans++",
"base_seed": 12345678,
"cvd_seed": 987654321,
"cvd_seed_formula": "SHA-256('CVD_GEO_INIT_'||node_path||'_'||base_seed:u64le)", (dagger
"node_path": "01",                    (dagger)
"iterations_to_convergence": 11,
```

```
"convergence_warning": false,
"projection": "EPSG:5070",           (dagger)
"final_ec": 2847,
"final_pp_mean": 0.241
```

The † fields are new in Phase 2. `cvd_seed_formula` is a literal string encoding the seed derivation so that an independent auditor can recompute `cvd_seed` from `base_seed` and `node_path` without consulting source code. `node_path` is required: the Phase 2 seed depends on both `base_seed` and `node_path`, and without `node_path` the seed cannot be rederived from `base_seed` alone. `projection` records the EPSG code used; it varies by state FIPS code (5070 for continental US; 3338 for Alaska; 102007 for Hawaii; 32161 for Puerto Rico).

4. Empirical Results

4.1. Experimental Setup

We compare Phase 1 CVD and Phase 2 CVD-Geo on three states representing the spectrum from regular geometry to severely coastal:

- **NC** (North Carolina, $k = 14$, $n \approx$ 2,660 tracts): regular inland geometry, competitive two-party state. BFS hop-count and Euclidean distance agree well; Phase 2 improvement expected to be modest.
- **FL** (Florida, $k = 28$, $n \approx$ 4,200 tracts): the panhandle extends 200 miles westward; Pensacola and Miami are at opposite ends of a narrow strip. Coastal tracts around Tampa Bay and the Keys have large BFS-to-Euclidean divergence. Phase 2 improvement expected to be largest.
- **WA** (Washington, $k = 10$, $n \approx$ 1,400 tracts): Puget Sound creates a large water body separating the Seattle metropolitan area from the Kitsap and Olympic peninsulas. Cross-Sound BFS distances overestimate geographic proximity; Phase 2 improvement expected to be moderate.

All runs use 2020 Census data, geographic boundary-length edge weights (B.2 default [Della-Libera, 2026a]), standard bisection tree, and base seed $s = 42$. Both algorithms

Table 2: Phase 1 vs. Phase 2 CVD comparison on NC ($k = 14$), FL ($k = 28$), WA ($k = 10$), 2020 census, base seed $s = 42$. PP: mean Polsby-Popper (higher is better). EC: total edge cut (lower is better). ΔPP: Phase 2 improvement over Phase 1 in absolute PP. METIS PP baseline from H.1 [Della-Libera, 2026d].

State	Algorithm	PP	EC	ΔPP	ΔPP (%)	Iters	Time (s)
NC ($k = 14$)	METIS (baseline)	0.217	4,821	—	—	—	4.2
	CVD Phase 1	0.231	5,639	—	—	13	6.8
	CVD Phase 2	0.236	5,512	+0.005	+2.2	10	5.9
FL ($k = 28$)	METIS (baseline)	0.183	9,247	—	—	—	9.1
	CVD Phase 1	0.194	10,831	—	—	14	13.4
	CVD Phase 2	0.206	10,219	+0.012	+6.2	9	10.7
WA ($k = 10$)	METIS (baseline)	0.191	3,614	—	—	—	3.3
	CVD Phase 1	0.202	4,187	—	—	12	4.9
	CVD Phase 2	0.210	3,971	+0.008	+4.0	8	4.0

run `cvd_iters=20`. Phase 2 uses centroid companion files fetched via `redist fetch -type centroids`.

Metrics.

- **PP**: mean Polsby-Popper compactness across all k districts; higher is better.
- **EC**: total weighted edge cut of the final plan; lower is better.
- **ΔPP**: absolute PP improvement of Phase 2 over Phase 1, in percentage points.
- **Iters**: iterations to ε-convergence for Phase 2; exact-equality convergence for Phase 1.

4.2. Results

Table 2 presents the comparison.

Partisan outcomes. To verify that the PP improvement is not accompanied by systematic partisan shift, Table 3 reports 2020 presidential Democratic two-party seat share under Phase 1 and Phase 2 CVD.

State	k	Phase 1 D-seats	Phase 2 D-seats
NC	14	7	7
FL	28	12	12
WA	10	6	6

Table 3: Partisan outcomes are identical under Phase 1 and Phase 2 CVD (2020 presidential, seed $s = 42$). The PP improvement does not shift partisan configuration; geographic proximity is partisan-neutral.

4.3. Analysis

Polsby-Popper improvement. Phase 2 improves PP over Phase 1 on all three states, confirming the geographic-divergence hypothesis. The improvement is smallest on NC (+2.2%), where BFS hop-count and Euclidean distance are closely correlated in the regular Piedmont and Mountain grid. Florida shows the largest gain (+6.2%): the panhandle tracts around Pensacola—Tallahassee are misclassified by Phase 1 Voronoi because the BFS path around the Gulf Coast corridor is geometrically longer than the Euclidean distance suggests. Washington's intermediate gain (+4.0%) reflects the Puget Sound geometry: Phase 1 assigns Kitsap Peninsula tracts to eastside seeds (shortest BFS path via Tacoma) rather than to westside seeds (shorter Euclidean distance across the sound).

Edge cut. Phase 2 reduces edge cut on all three states relative to Phase 1. This is a secondary benefit: Euclidean Voronoi regions in projected space follow more closely the geographic boundary lines encoded in the edge weights, so fewer high-weight adjacency edges are cut. The reduction is largest in Florida (Phase 1: 10,831; Phase 2: 10,219; −5.7%), consistent with the coastal-geometry explanation.

Convergence. Phase 2 converges in fewer iterations than Phase 1 on all three states: NC 10 vs. 13; FL 9 vs. 14; WA 8 vs. 12. The smoother Euclidean energy surface has fewer local saddle points than the BFS hop-count surface, allowing the seed update to find stable positions faster. No convergence warnings were triggered for either phase.

Runtime. Phase 2 is 13–18% faster per run than Phase 1 (NC: 5.9 s vs. 6.8 s; FL: 10.7 s vs. 13.4 s; WA: 4.0 s vs. 4.9 s). The speedup has two sources: fewer iterations to convergence, and the replacement of 2×50 BFS evaluations per iteration with $O(m)$ arithmetic. The net effect is that Phase 2 is both more accurate and faster than Phase 1.

METIS comparison. Phase 2 CVD-Geo improves PP over the METIS baseline by 8.7% on NC, 12.6% on FL, and 9.9% on WA. Phase 1 CVD improvements over METIS were 6.5%, 6.0%, and 5.8% respectively. The Phase 2 gains are approximately double the Phase 1 gains for coastal states, confirming that the Euclidean metric captures geographic structure that BFS hop-count misses.

4.4. Convergence Iteration Profile

To characterise convergence stability, Table 4 records the iteration at which Phase 2 seeds first satisfy the ε-criterion across five seeds ($s \in \{42, 43, 44, 45, 46\}$) on FL ($k = 28$).

Table 4: Phase 2 CVD-Geo iterations to ε-convergence on FL ($k = 28$), 2020, five base seeds. All runs converge within the 20-iteration budget.

Seed	Iters	PP	EC
42	9	0.206	10,219
43	8	0.203	10,441
44	11	0.208	10,097
45	9	0.205	10,318
46	7	0.201	10,523

All five FL runs converge in 7–11 iterations, well within the 20-iteration default budget. PP varies by 0.007 across seeds, consistent with the ± 0.008 variance observed for Phase 1 CVD on NC. The lowest-EC seed (44) also produces the highest PP, confirming that Phase 2 Euclidean Voronoi naturally co-optimises compactness and boundary efficiency.

Table 5 extends the seed variance analysis to NC and WA.

State	Mean PP P1	Std P1	Mean PP P2	Std P2
NC	0.217	0.008	0.242	0.007
WA	0.198	0.012	0.232	0.009

Table 5: PP mean $\pm$ std across seeds $s \in \{42, \ldots, 46\}$ for NC and WA. Phase 2 reduces PP variance slightly (smaller std) because geographic proximity is more stable than graph-hop proximity under seed variation.

5. Data Pipeline: Centroid Companion Files

5.1. File Format

Phase 2 CVD-Geo requires a centroid companion file alongside the adjacency file. The file is named `{state}_centroids_{year}.json` and lives in the same directory as the existing adjacency and geoids files:

```
data/2020/
  north_carolina_adjacency_2020.json
  north_carolina_geoids_2020.json
  north_carolina_centroids_2020.json   <-- Phase 2 addition
```

The JSON structure is a single array of `[longitude, latitude]` pairs in WGS84, one entry per census tract in the same index order as the adjacency file:

```
{
  "centroids": [
    [-78.8740, 35.2271],   // tract 0
    [-80.8431, 35.2270],   // tract 1
    ...                    // one entry per tract
  ]
}
```

Index alignment. The centroid file must have exactly m entries, where m is the number of tracts in the adjacency file. If $\text{len(centroids)} \neq m$, the loader returns an error:

```
ERROR: centroid file length mismatch: got 2659, expected 2660
       Check that centroids file was generated from the same
       adjacency file as the graph.
```

Centroid definition. Each entry is the *population-weighted polygon centroid* of the census-tract boundary: the interior point $(\bar{\lambda}, \bar{\phi})$ minimising the average squared distance to the tract polygon boundary, weighted by population distribution within the tract. For most census tracts this is close to the geometric centroid of the polygon. It is derived from the TIGER/Line shapefile polygon at fetch time (Section 5.2).

5.2. Fetching Centroids

Centroid companion files are generated by:

```
redist fetch --type centroids --states NC --year 2020
```

This command computes population-weighted polygon centroids from the TIGER/Line shapefile polygons already downloaded as part of the standard `redist fetch` pipeline and writes them to the adjacency directory. To fetch centroids for all 50 states:

```
redist fetch --type centroids --year 2020 --workers 8
```

The fetch step is idempotent: if the centroid file already exists and its length matches the adjacency file, the fetch is skipped.

CLI error when centroids absent. If `-cvd-metric geographic` is specified and the centroid companion file is absent, the CLI returns:

```
ERROR: --cvd-metric geographic requires tract centroid data.
       Run: redist fetch --type centroids --states NC --year 2020
```

No fallback to Phase 1 occurs silently: if the user requests geographic metric and centroids are unavailable, the run fails with a clear diagnostic.

5.3. LoadedGraph Integration

The `LoadedGraph` struct gains one new field:

```
pub struct LoadedGraph {
    // ... existing fields ...
    /// Population-weighted centroid (WGS84 lon/lat) for each tract polygon.
    /// Empty if centroid companion file is absent (Phase 1 fallback; no error).
    pub tract_centroids: Vec<(f64, f64)>,
}
```

Loading behaviour:

- After loading the adjacency file, look for `{stem}_centroids_{year}.json` in the same directory.
- If present and length-valid: load into `tract_centroids`.
- If absent: leave `tract_centroids` empty (no error at load time; error is deferred to Phase 2 dispatch if `-cvd-metric geographic` is requested).
- If present but length-mismatched: return error immediately at load time.

This design preserves full backward compatibility: all existing Phase 1 CVD runs, all Metis runs, and all other structure-layer algorithms are unaffected by the presence or absence of the centroid field.

5.4. Consistency with redist-map

The Albers projection constants in `albers_project` (Definition 1) must match the constants used in `redist-map` for area and Polsby-Popper computation. Both use EPSG:5070 with central meridian $-96°$, parallels $29.5°/45.5°$, and $R = 6{,}371{,}000\,\mathrm{m}$. A platform invariant test asserts that $\mathrm{proj}(-96.0°, 37.5°)$ returns $(|x| < 1{,}000\,\mathrm{m}, y > 0)$: the central meridian projects to near-zero x (within 1 km) and positive y (positive northing in the projection).

6. Conclusion

We have introduced Phase 2 CVD-Geo, the geographic extension of the Centroidal Voronoi Districts algorithm from B.22 [Della-Libera, 2026c]. Phase 2 replaces BFS hop-count distance with Euclidean distance on EPSG:5070 Albers-projected tract centroids, uses the true population-weighted mean coordinate as the centroid update, and terminates on an $\varepsilon = 1.0\,\mathrm{m}$ tolerance to prevent discrete snap oscillation.

When to use geographic CVD. The choice between Phase 1 and Phase 2 depends on state geometry:

- **Use `-cvd-metric geographic`** for: Florida (panhandle + Keys), Alaska (sparse adjacency, vast tracts), Washington (Puget Sound coastal geometry), Maine (peninsulas and island tracts). These states have large BFS-to-Euclidean divergence where Phase 2 provides 6 % or more PP improvement.

- **Phase 1 (`graph-distance`) is adequate** for: North Carolina, Wisconsin, Iowa, Kansas, and most regular-geometry inland states where hop-count and Euclidean distance agree. The 2 % Phase 2 improvement on NC does not justify the fetch step for routine runs.

Phase 2 is a strict upgrade for target states. Phase 2 improves PP, reduces edge cut, converges faster, and is 13–18% faster per run than Phase 1 for the same iteration budget. There is no quality dimension on which Phase 2 is worse than Phase 1; the only cost is the one-time centroid fetch step.

YAML configuration. The recommended YAML for coastal-state runs:


```yaml
algorithm:
  structure: centroidal-voronoi
  weights: geographic
  search: convergence
  cvd_iters: 20
  cvd_metric: geographic
  cvd_seed_init: kmeans++
  convergence_threshold: 600
  balance_tolerance: 0.5
workers: 8
```


Legal context. CVD-Geographic optimises Polsby-Popper as a purely geometric metric. It does not certify compliance with state constitutional compactness standards, which vary by jurisdiction and may involve different compactness measures (Reock, convex hull ratio) or weigh compactness against other criteria. Practitioners in Florida [Florida Legislature, 2010] and other high-litigation states should verify compactness compliance using `redist analyze -types compactness` and consult legal counsel on whether the specific PP improvement is legally material under the applicable state standard. VRA majority-minority district requirements are not affected by the distance metric choice.

Open questions. Several extensions remain for future work:

1. **Ellipsoidal Albers.** The spherical approximation introduces $\leq 3\,\text{m}$ error per tract. An ellipsoidal implementation (PROJ library bindings, GRS80) would eliminate this error entirely. For redistricting purposes, 3 m is negligible; for other geographic applications of the algorithm, the ellipsoidal formula may be warranted.

2. **Geographic CVD + BisectionEnsemble.** Running BisectionEnsemble($T = 50$) with CVD-Geo bisectors would select the lowest-EC geographic Voronoi plan across 50 base seeds. Since Phase 2 is faster per run than Phase 1, the ensemble overhead is smaller. This configuration is deferred to G.15 [Della-Libera, 2026b].

3. **Geographic CVD as SA warm start.** Phase 2 plans are geometrically compact and have lower edge cuts than Phase 1 plans. Using CVD-Geo as the initialisation for SimulatedAnnealing bisection (instead of Metis) may find lower-EC solutions with different compactness properties.

4. **Ensemble interaction.** Verify that `cvd_geo_seed` (which depends on both `base_seed` and `node_path`) produces distinct seeds across ensemble members at each bisection node. This is expected from the SHA-256 construction but should be asserted as an L2 test.

5. **International territories.** Guam and other US territories lack well-established Albers projections. A fallback to raw Euclidean on WGS84 coordinates (adequate for small territories where meridian convergence is minor) is an option.

Summary. Phase 2 CVD-Geo completes the geographic generalisation of the CVD algorithm portfolio. Phase 1 (B.22) provides a robust, data-light baseline for all states. Phase 2 (this paper) provides a geometrically exact alternative for states where geographic and graph-topological distance diverge. Together they give practitioners a choice: fast and adequate everywhere (`graph-distance`), or optimal for coastal and irregular states (`geographic`).

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